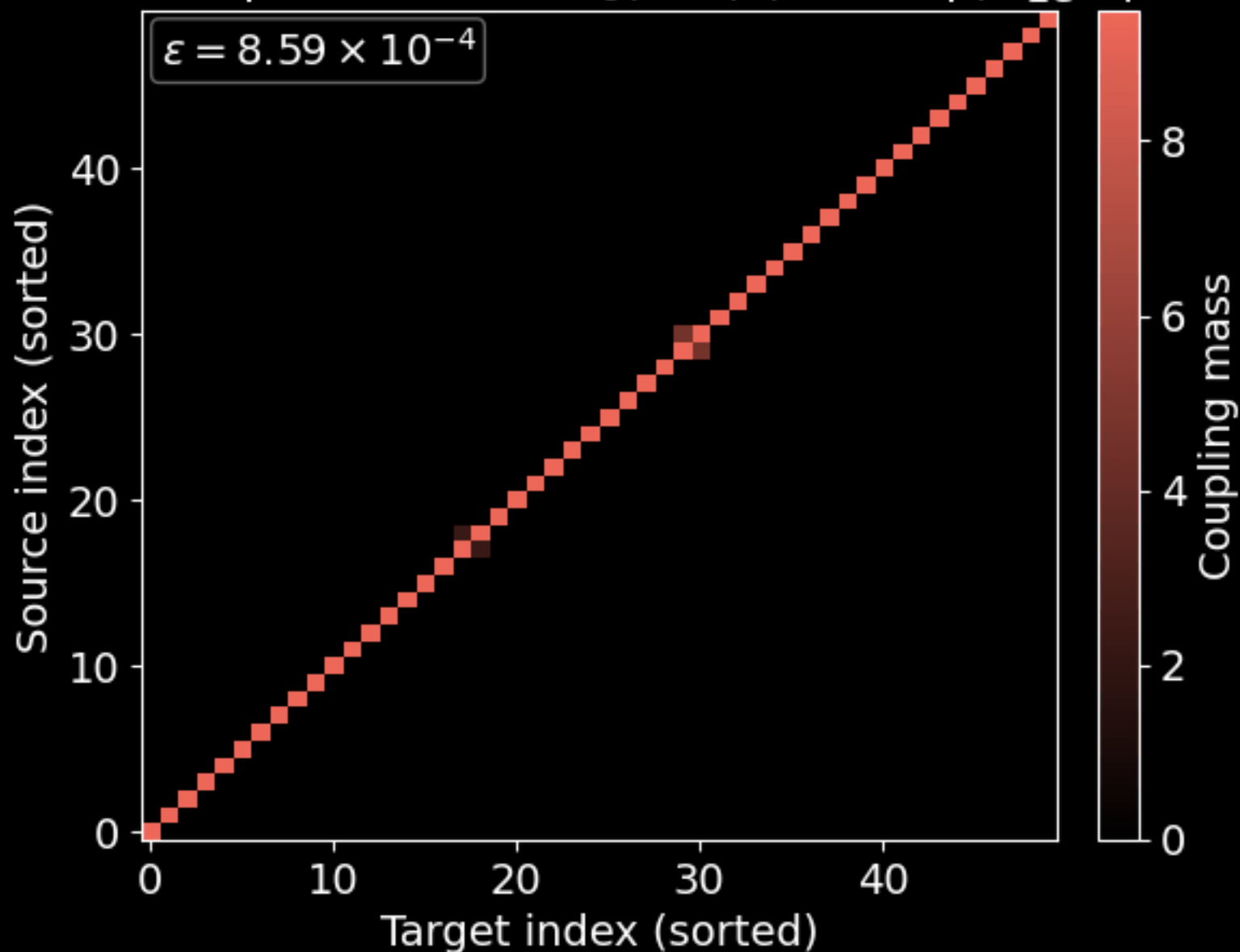


non-diagonal coupling

subtract self-costs to remove bias

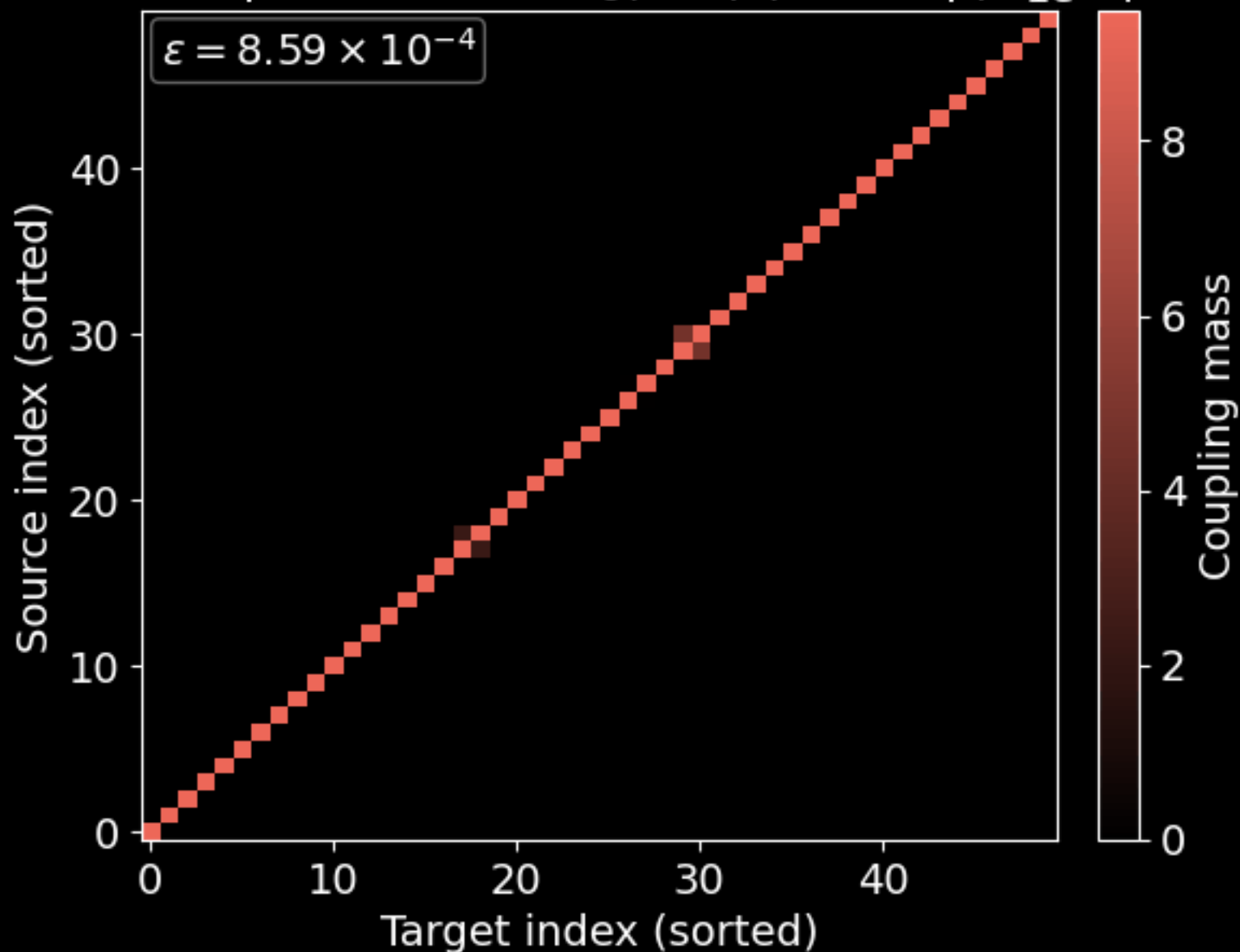
Optimal π for $\text{OT}_\varepsilon(\alpha, \alpha)$ (ε sweep) $1\text{e}-4$

$\varepsilon = 8.59 \times 10^{-4}$



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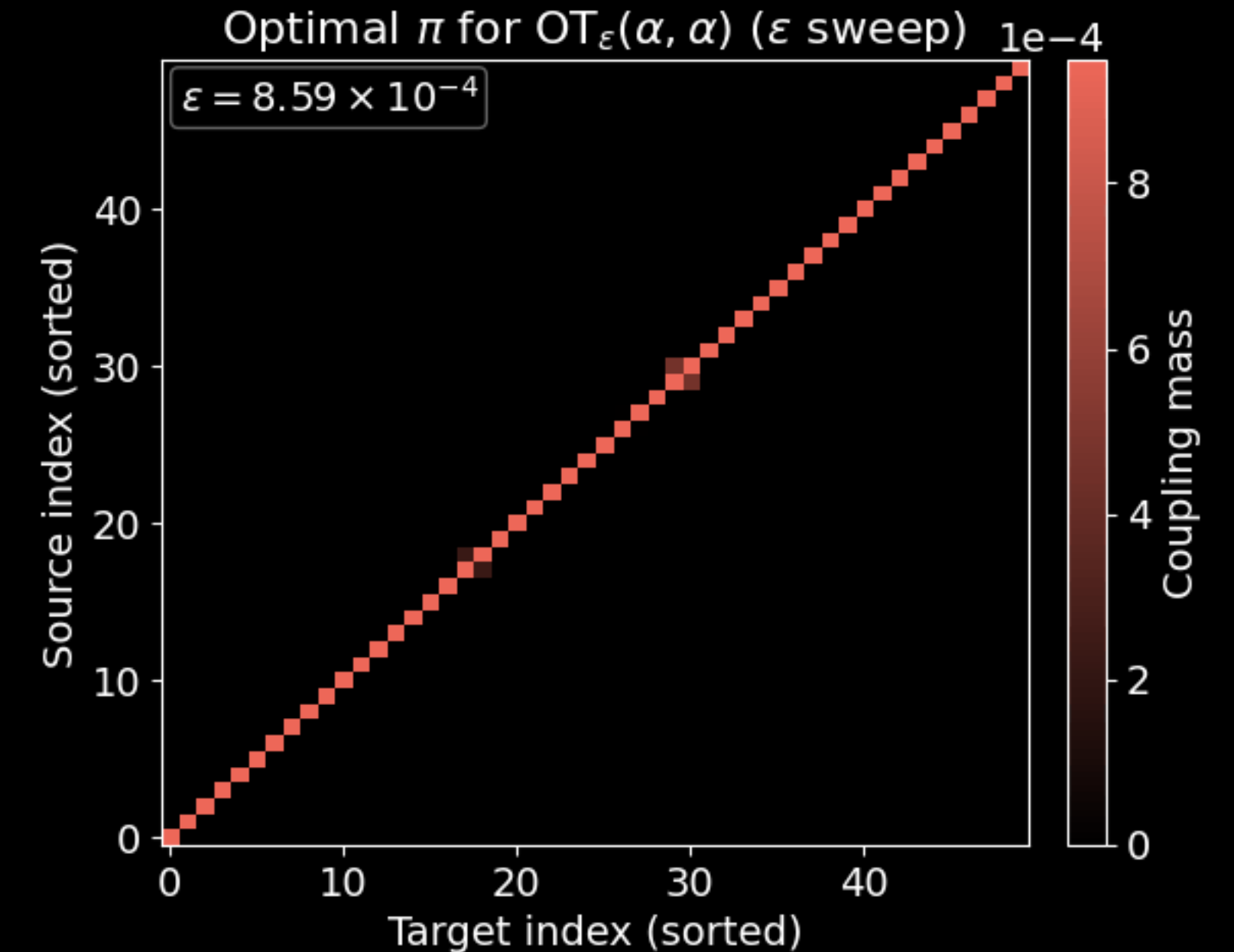
Debiasing Entropic OT

- Where does the bias come from?
- In $\text{OT}_\varepsilon(\alpha, \alpha) = \min_{\pi \in \Pi(\alpha, \alpha)} \langle C, \pi \rangle - \varepsilon H(\pi)$
- Second term induces **non-diagonal coupling**
- Both terms are non-zero!
- Idea: **subtract self-costs to remove bias**, e.g.

$$\text{OT}_\varepsilon(\alpha, \nu) - \text{OT}_\varepsilon(\alpha, \alpha) = 0 \quad \text{if } \nu = \alpha$$

$$\text{OT}_\varepsilon(\nu, \beta) - \text{OT}_\varepsilon(\beta, \beta) = 0 \quad \text{if } \nu = \beta$$

- We need both, so take $\text{OT}_\varepsilon(\alpha, \beta) - \frac{1}{2}\text{OT}_\varepsilon(\alpha, \alpha) - \frac{1}{2}\text{OT}_\varepsilon(\beta, \beta)$



Sinkhorn Divergence

- **Sinkhorn Divergence** [Ramdas et al. 2017; Salimans et al. 2018; Genevay et al. 2018]: